Prathmesh Sayal

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Experience

Summer Intern, Fidelity Investments - Bangalore May 2026 – Jul 2026

- Built a reusable test automation framework covering 2 extraction and validation scenarios, centralizing result persistence and improving data-pipeline test coverage by 10%
- Automated validation and reconciliation of data extracted from Azure Blob Storage against source records, ensuring high data accuracy and consistency across thousands of participant records

Project Intern, Hewlett Packard Enterprise(HPE)-Remote Apr 2025 – Jan 2026

- Architected a Confidential Computing prototype on AWS & Kubernetes to run workloads in secure enclaves, eliminating 3 vulnerabilities flagged across penetration tests
- Estimated resource requirements and technical effort for custom extensibility modules across 5 product teams to align with enterprise business goals

Intern Trainee, ISRO-Leos Bangalore Aug 2025 – Sep 2025

- Engineered an AI pipeline to process and interpret satellite imagery, enhancing object recognition and enabling sub-meter precision in autonomous navigation control systems with 10 seconds training time
- Designed and prototyped novel ML algorithms for spacecraft simulation environments, achieving $\tilde{8}0\%$ accuracy and low-latency performance for real-time autonomous maneuvering

Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript
Frameworks: React.js, Node.js, Next.js, PyTorch, TensorFlow, RESTful API
Methodologies & Automation: Agile, Scrum, SDLC, Low-code Development, Workflow Automation, Intelligent Automation, System Integration
Tools & Platforms: SQL, PostgreSQL, MySQL, GCP, Docker, Kubernetes, MongoDB, GraphQL, Git, GitHub, Generative AI, Windows, Unix

Projects

RankTransport Index: Dynamic Learned Index for Hierarchical String Data *GitHub*

- Developed and implemented a Python/C++ dynamic learned index for hierarchical string keys, using sparse-radix lexicodes and rank transport to keep max rank error at 64 while stale learned predictions drifted up to 681 across six workloads
- Established a reproducible benchmark suite with public corpora and pinned baselines including ALEX, PGM, LIPP, ART, and HOT; measured up to 8.68M lookups/sec with 115 ns average lookup latency on URL-path workloads

CortexMD (Team Lead, Full Stack Developer) *GitHub*

- Implemented Explainable AI pipelines using first-order logic, knowledge graphs, and confidence-based medical verification across multiple diagnostic pathways for transparent decision-making
- Delivered clinical-ready modules with UMLS integration, ontology mapping, and NVIDIA Clara-powered imaging support for 3 imaging modalities

Fence.AI - Deepfake Detection and Media Verification (Backend Developer) *GitHub*

- Created an XAI-powered deepfake detection and protection service in Python, authenticating media across 2 social platforms by Meta(Whatsapp and Instagram Integration) before upload
- Filed a patent on the technology to protect media through 3-phase layer aberration injection that hallucinates the image-generation model

Education

Ramaiah Institute of Technology, B.E in Computer Science Engineering Aug 2023 – May 2027
Specialisation in **Cybersecurity**

- CGPA: 9.52/10.00 (Till 6th Semester)
- **Coursework:** Computer Organisation Architecture, Computer Networking, Computational Theory, Compiler Design, DataBase Management System, Operating System, DevOps, RDBMS, MERN Stack, NoSQL, SAML, Cryptography, Machine Learning

Publications

Quantum Powered Data Analysis — ICEMTEC Pune 2024

- Published a paper exploring the intersection of quantum computing and data analytics

Achievements

- Runner-Up, Smart India Hackathon 2024 (EDUMITRA) — National-level innovation challenge, organised under the guidance of the Hon'ble Prime Minister of India
- Winner, HPE Think-a-thon 2025 — Brainstormed 10 DS Problems under time constraint environment
- Winner, CIDECODE PES 2025 — Fence.AI project recognised by the Hon'ble Home Minister of Karnataka
- 10x National Level Hackathon Winner and 7x Capture the Flag Winner

Extra-Curricular

EXECOM, IEEE RIT-B - Computational Intelligence Society Jan 2025 – Jan 2026
Lead, Google Developer Group (GDG) – RIT Dec 2024 – Present